

APPM 4600 — HOMEWORK # 4

1. Let $f(x)$ denote a function with root α of multiplicity $m > 1$.
 - (a) Write down a formal mathematical definition of what it means for α to be a root of multiplicity m of $f(x)$.
 - (b) Show that Newton's method applied to $f(x)$ only converges linearly to the root α .
 - (c) Show that the fixed point iteration applied to $g(x) = x - m \frac{f(x)}{f'(x)}$ is second order convergent.
 - (d) What does part (c) provide for Newton's method in the case of roots with multiplicity greater than 1?
2. Beginning with the definition of order of convergence of a sequence $\{x_k\}_{k=1}^{\infty}$ that converges to α , derive a relationship between the $\log(|x_{k+1} - \alpha|)$ and $\log(|x_k - \alpha|)$. What is the order p in this relationship?
3. There are two ways of improving the convergence of Newton's method when a root has multiplicity greater than 1: Problem 1 c and apply Newton's method to $g(x) = \frac{f(x)}{f'(x)}$.

In this problem consider finding the root of the function $f(x) = e^{3x} - 27x^6 + 27x^4e^x - 9x^2e^{2x}$ in the interval $(3, 5)$.

Explore the order of convergence when applying (i) Newton's method, (ii) the modified Newton's method from class and (iii) the modified Newton's method in Problem 1. Which method do you prefer and why?
4. Use Newton and Secant method to approximate the largest root of

$$f(x) = x^6 - x - 1.$$

Start Newton's method with $x_0 = 2$. Start Secant method with $x_0 = 2$ and $x_1 = 1$.

- (a) Create a table of the error for each step in the iteration. Does the error decrease as you expect?
- (b) Plot $|x_{k+1} - \alpha|$ vs $|x_k - \alpha|$ on log-log axes where α is the exact root for both methods. What are the slopes of the lines that result from this plot? How does this relate to the order?